

MULTI-AGENT SERVING: WORKLOADS → OPTIMIZATION FAMILIES

Workload characteristic	Memory optimization	Prefix-aware scheduling	Cache lifecycle	Power management
Sparse activation few agents generate at once	ScaleSim prefetch / eviction	Activity-aware placement	Idle KV reclamation	KAIROS phase-aware power
Invocation distance predictable next-use gap	ScaleSim distance-aware eviction	TOPAS / TokenCake future-aware placement	Continuum TTL / resume policy	Idle phases expose slack
Inter-agent sharing shared prompts / KV prefixes	Shared-state memory pressure	TokenCake / KVFlow prefix-aware routing	ForkKV / KVCOMM reuse / copy-on-write	Indirect efficiency effect

Current gap: surveyed systems optimize subsets of the stack; diagnostic signals are not a first-class serving input.